Method of Differentiation: Questions and Full Solutions

Question 01

English

Let C be the curve given by $x = t\log_e t,\ y = t^2,\ t > 0$. Let P be the point on C at which the tangent to C passes through the origin O. The normal to C at P meets the x-axis at the point N. Then the area of the triangle OPN is equal to

(A) $\dfrac{e^2\left(1+e^2\right)}{2}$

(B) $\dfrac{e^3\left(e^2-1\right)}{2}$

(C) $\dfrac{e^3\left(1+e^2\right)}{2}$

(D) $e^3\left(1+e^2\right)$

Only one option is correct.

हिन्दी

माना कि C वक्र $x = t\log_e t,\ y = t^2,\ t > 0$ द्वारा दिया गया है। माना कि P वक्र C पर वह बिन्दु है जिस पर C की स्पर्श रेखा मूलबिन्दु O से होकर जाती है। P पर C का अभिलम्ब x-अक्ष को बिन्दु N पर मिलता है। तब त्रिभुज OPN का क्षेत्रफल बराबर है

(A) $\dfrac{e^2\left(1+e^2\right)}{2}$

(B) $\dfrac{e^3\left(e^2-1\right)}{2}$

(C) $\dfrac{e^3\left(1+e^2\right)}{2}$

(D) $e^3\left(1+e^2\right)$

केवल एक विकल्प सही है।

Solution

For the parametric curve, $\dfrac{dx}{dt} = \log_e t + 1$ and $\dfrac{dy}{dt} = 2t$. Where $\log_e t + 1 \neq 0$, $\dfrac{dy}{dx} = \dfrac{2t}{1+\log_e t}$. (At $t = \dfrac{1}{e}$ the tangent is the vertical line $x = -\dfrac{1}{e}$, which does not pass through O.)

The tangent at the point with parameter t passes through O exactly when the slope of OP equals the slope of the tangent. At $t = 1$ the point is $(0, 1)$, where the tangent is $y = 2x + 1$; it does not pass through O. For $t \neq 1$, $x \neq 0$ and the slope of OP is $\dfrac{y}{x} = \dfrac{t^2}{t\log_e t} = \dfrac{t}{\log_e t}$.

Equating: $\dfrac{2t}{1+\log_e t} = \dfrac{t}{\log_e t}$. Since $t > 0$, cancel t: $2\log_e t = 1 + \log_e t$, so $\log_e t = 1$ and $t = e$. Hence P is unique: $P = (e\log_e e,\ e^2) = (e, e^2)$.

Slope of the tangent at P: $\dfrac{2e}{1+1} = e$ (the tangent is $y = ex$, which indeed passes through O). Slope of the normal: $-\dfrac{1}{e}$.

Normal at P: $y - e^2 = -\dfrac{1}{e}(x - e)$. Putting $y = 0$: $x - e = e^3$, so $N = (e + e^3, 0)$.

Triangle OPN has base $ON = e + e^3$ on the x-axis and height equal to the ordinate of P, namely e^2. Area $= \dfrac{1}{2}\left(e + e^3\right)e^2 = \dfrac{e^3\left(1+e^2\right)}{2}$.

(A sign slip in the normal gives $N = (e - e^3, 0)$ and option (B); omitting the factor $\frac{1}{2}$ gives (D).)

Correct Option: (C)

Question 02

English

Let C be the curve given by $x = t^3 + 3t,\ y = t^4 + 2t^2,\ t \in \mathbb{R}$. Let P be the point on C at which the tangent to C is parallel to the line $4x - 3y + 5 = 0$. Then the value of $\frac{d^3y}{dx^3}$ at P is equal to

(A) $-\frac{1}{27}$

(B) $-\frac{2}{9}$

(C) $\frac{1}{27}$

(D) $-\frac{2}{27}$

Only one option is correct.

हिन्दी

माना कि C वक्र $x = t^3 + 3t,\ y = t^4 + 2t^2,\ t \in \mathbb{R}$ द्वारा दिया गया है। माना कि P वक्र C पर वह बिन्दु है जिस पर C की स्पर्श रेखा, रेखा $4x - 3y + 5 = 0$ के समान्तर है। तब P पर $\frac{d^3y}{dx^3}$ का मान बराबर है

(A) $-\frac{1}{27}$

(B) $-\frac{2}{9}$

(C) $\frac{1}{27}$

(D) $-\frac{2}{27}$

केवल एक विकल्प सही है।

Solution

$\frac{dx}{dt} = 3t^2 + 3 = 3(t^2 + 1)$, which is positive for all t, so x is strictly increasing in t and y is a differentiable function of x along the whole curve.

$\frac{dy}{dt} = 4t^3 + 4t = 4t(t^2 + 1)$. Hence $\frac{dy}{dx} = \frac{4t(t^2+1)}{3(t^2+1)} = \frac{4t}{3}$.

The line $4x - 3y + 5 = 0$ has slope $\frac{4}{3}$. The tangent is parallel to it when $\frac{4t}{3} = \frac{4}{3}$, i.e. $t = 1$. Then $P = (1 + 3,\ 1 + 2) = (4, 3)$, and P is unique.

For higher derivatives with respect to x, differentiate with respect to t and divide by $\frac{dx}{dt}$ each time:

$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{4t}{3}\right)}{3(t^2+1)} = \frac{4}{9(t^2+1)}$.

$\frac{d}{dt}\left(\frac{4}{9(t^2+1)}\right) = -\frac{8t}{9(t^2+1)^2}$, so $\frac{d^3y}{dx^3} = -\frac{8t}{9(t^2+1)^2} \cdot \frac{1}{3(t^2+1)} = -\frac{8t}{27(t^2+1)^3}$.

At $t = 1$: $\frac{d^3y}{dx^3} = -\frac{8}{27 \cdot 8} = -\frac{1}{27}$.

(Forgetting the division by $\frac{dx}{dt}$ in the last step gives $-\frac{8}{36} = -\frac{2}{9}$, option (B).)

Correct Option: (A)

Question 03

English

Let $L = \lim\limits_{x\to 2} \frac{\left(x^3-7\right)^{x^x}-1}{(x-1)^x-1}$. Then the value of L is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $L = \lim\limits_{x\to 2} \frac{\left(x^3-7\right)^{x^x}-1}{(x-1)^x-1}$ है। तब L का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $N(x) = \left(x^3-7\right)^{x^x} = e^{x^x \log_e\left(x^3-7\right)}$ and $D(x) = (x-1)^x = e^{x\log_e(x-1)}$, defined near $x = 2$ (there $x^3 - 7 > 0$ and $x - 1 > 0$). At $x = 2$: $x^3 - 7 = 1$ and $x - 1 = 1$, so $N(2) = D(2) = 1$ and L is a $\frac{0}{0}$ form: $L = \lim\limits_{x\to 2} \frac{N(x)-N(2)}{D(x)-D(2)} = \frac{N'(2)}{D'(2)}$, provided $D'(2) \neq 0$.

Logarithmic differentiation of N: $\log_e N = x^x \log_e\left(x^3-7\right)$, so $\frac{N'}{N} = (x^x)' \log_e\left(x^3-7\right) + x^x \cdot \frac{3x^2}{x^3-7}$, where $(x^x)' = x^x\,(1+\log_e x)$.

At $x = 2$: $\log_e\left(x^3-7\right) = \log_e 1 = 0$, so the first term vanishes whatever $(x^x)'$ is. The second term is $2^2 \cdot \frac{12}{1} = 48$. Hence $N'(2) = N(2) \cdot 48 = 48$.

Logarithmic differentiation of D: $\log_e D = x\log_e(x-1)$, so $\frac{D'}{D} = \log_e(x-1) + \frac{x}{x-1}$. At $x = 2$ this is $0 + 2 = 2$, so $D'(2) = 2 \neq 0$.

Independent check with standard limits: as $x \to 2$, $e^u - 1 \sim u$ when $u \to 0$. Here $u_N = x^x \log_e\left(x^3-7\right)$ and $\log_e\left(1+(x^3-8)\right) \sim x^3 - 8 \sim 12(x-2)$, so the numerator $\sim 4 \cdot 12(x-2) = 48(x-2)$. Similarly $u_D = x\log_e\left(1+(x-2)\right) \sim 2(x-2)$.

Therefore $L = \frac{48(x-2)}{2(x-2)} \to 24$.

Final Answer: 24

Question 04

English

Let $I \subseteq (0, \infty)$ be an open interval containing 1, and let $y : I \to (0, \infty)$ be a differentiable function such that $y(1) = 1$ and $x^y + 3y^x = 4xy^2$ for all $x \in I$, where $y = y(x)$. Then the value of the derivative of y^x with respect to x^y at $x = 1$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $I \subseteq (0, \infty)$ एक विवृत अन्तराल है जिसमें 1 है, तथा $y : I \to (0, \infty)$ एक अवकलनीय फलन है जिसके लिए $y(1) = 1$ तथा सभी $x \in I$ के लिए $x^y + 3y^x = 4xy^2$ है, जहाँ $y = y(x)$ है। तब $x = 1$ पर x^y के सापेक्ष y^x के अवकलज का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $u = x^y$ and $v = y^x$, both functions of x on I. The derivative of v with respect to u is $\dfrac{dv}{du} = \dfrac{dv/dx}{du/dx}$ wherever $\dfrac{du}{dx} \neq 0$.

Logarithmic differentiation of u: $\log_e u = y \log_e x$, so $\dfrac{1}{u}\dfrac{du}{dx} = y' \log_e x + \dfrac{y}{x}$, i.e. $\dfrac{du}{dx} = x^y \left(y' \log_e x + \dfrac{y}{x} \right)$.

Logarithmic differentiation of v: $\log_e v = x \log_e y$, so $\dfrac{dv}{dx} = y^x \left(\log_e y + \dfrac{x\,y'}{y} \right)$.

At $x = 1$, $y = 1$: $x^y = y^x = 1$ and $\log_e x = \log_e y = 0$. Hence $\dfrac{du}{dx} = 1 \neq 0$ and $\dfrac{dv}{dx} = y'(1)$.

Implicit differentiation of $u + 3v = 4xy^2$: $\dfrac{du}{dx} + 3\dfrac{dv}{dx} = 4y^2 + 8xyy'$. At $(1, 1)$: $1 + 3y'(1) = 4 + 8y'(1)$, so $5y'(1) = -3$ and $y'(1) = -\dfrac{3}{5}$.

(The point is consistent: at $x = 1$ the relation reads $1 + 3y = 4y^2$, whose positive root is $y = 1$; the derivative of the relation with respect to y there is $3 - 8 = -5 \neq 0$, so $y'(1)$ is determined.)

Therefore $\dfrac{dv}{du} = \dfrac{-3/5}{1} = -\dfrac{3}{5} = -0.60$.

Final Answer: -0.60

Question 05

English

Let C be the curve $y^2 + y = x^3 - x$ in the xy-plane, and let O be the origin. The tangent to C at O meets C again at the point Q. The tangent to C at Q meets C again at the point $R(\alpha, \beta)$. Then the value of $\alpha\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि xy-तल में C वक्र $y^2 + y = x^3 - x$ है तथा O मूलबिन्दु है। O पर C की स्पर्श रेखा C को पुनः बिन्दु Q पर मिलती है। Q पर C की स्पर्श रेखा C को पुनः बिन्दु $R(\alpha, \beta)$ पर मिलती है। तब $\alpha\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Differentiate $y^2 + y = x^3 - x$ implicitly with respect to x: $2y\dfrac{dy}{dx} + \dfrac{dy}{dx} = 3x^2 - 1$, so $\dfrac{dy}{dx} = \dfrac{3x^2 - 1}{2y + 1}$ wherever $2y + 1 \neq 0$.

At $O(0, 0)$ (which lies on C): $\dfrac{dy}{dx} = \dfrac{-1}{1} = -1$. The tangent at O is $y = -x$.

Points common to $y = -x$ and C: $x^2 - x = x^3 - x$, i.e. $x^3 - x^2 = 0$, i.e. $x^2(x - 1) = 0$. The double root $x = 0$ corresponds to the point of contact O; the other root gives $Q = (1, -1)$. (Check: $1 - 1 = 0 = 1 - 1$.)

At $Q(1, -1)$: $2y + 1 = -1$ and $3x^2 - 1 = 2$, so $\dfrac{dy}{dx} = -2$. The tangent at Q is $y + 1 = -2(x - 1)$, i.e. $y = 1 - 2x$.

Substituting in C: $(1-2x)^2 + (1-2x) = x^3 - x$, i.e. $4x^2 - 6x + 2 = x^3 - x$, i.e. $x^3 - 4x^2 + 5x - 2 = 0$. Since the line touches C at $x = 1$, $(x - 1)^2$ is a factor: $x^3 - 4x^2 + 5x - 2 = (x - 1)^2(x - 2)$.

The other point is $x = 2$, $y = 1 - 4 = -3$; check: $(-3)^2 + (-3) = 6$ and $2^3 - 2 = 6$. So $R = (2, -3)$, $\alpha = 2$, $\beta = -3$.

Hence $\alpha\beta = -6$.

Final Answer: -6

Question 06

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2 e^{-4x}$, and let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = f^{(9)}(x)$, where $f^{(9)}$ denotes the ninth derivative of f. Then the sum of the x-coordinates of all the points of inflection of the graph of $y = g(x)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 e^{-4x}$ द्वारा परिभाषित है, तथा $g : \mathbb{R} \to \mathbb{R}$, $g(x) = f^{(9)}(x)$ द्वारा परिभाषित है, जहाँ $f^{(9)}$, f का नौवाँ अवकलज है। तब $y = g(x)$ के आलेख के सभी नति परिवर्तन बिन्दुओं के x-निर्देशांकों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the Leibniz rule, $f^{(n)} = \sum_{k=0}^{n} \binom{n}{k} \left(x^2\right)^{(k)} \left(e^{-4x}\right)^{(n-k)}$. Only $k = 0, 1, 2$ contribute because $\left(x^2\right)^{(k)} = 0$ for $k \geq 3$, and $\left(e^{-4x}\right)^{(m)} = (-4)^m e^{-4x}$.

So $f^{(n)}(x) = e^{-4x}\left[(-4)^n x^2 + n(-4)^{n-1} \cdot 2x + \frac{n(n-1)}{2}(-4)^{n-2} \cdot 2\right] = (-4)^{n-2} e^{-4x}\left[16x^2 - 8nx + n(n-1)\right]$.

$g = f^{(9)}$ is twice differentiable and $g'' = f^{(11)}$. A point of inflection of $y = g(x)$ is where g'' changes sign.

With $n = 11$: $g''(x) = (-4)^9 e^{-4x}\left[16x^2 - 88x + 110\right]$. Since $e^{-4x} > 0$ and $(-4)^9$ is a non-zero constant, the sign of g'' changes exactly where $16x^2 - 88x + 110$ changes sign.

Its discriminant is $88^2 - 4 \cdot 16 \cdot 110 = 7744 - 7040 = 704 > 0$, so it has two distinct real roots $x = \frac{11 \pm \sqrt{11}}{4}$. At each simple root the quadratic changes sign, so g has exactly two points of inflection.

By Vieta, the sum of their x-coordinates is $\frac{88}{16} = \frac{11}{2} = 5.50$.

Final Answer: 5.50

Question 07

English

Let $F : (-1, 1) \to \mathbb{R}$ be defined by

$$F(x) = \begin{vmatrix} \sqrt{1+x} & \sin x & \log_e(1+x) \\ \tan x & e^{\sin x} & x^2 \\ 1-\cos x & e^x - 1 & \dfrac{1}{1+x} \end{vmatrix}$$

Then the value of $F'(0)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : (-1, 1) \to \mathbb{R}$ निम्न द्वारा परिभाषित है

$$F(x) = \begin{vmatrix} \sqrt{1+x} & \sin x & \log_e(1+x) \\ \tan x & e^{\sin x} & x^2 \\ 1-\cos x & e^x - 1 & \dfrac{1}{1+x} \end{vmatrix}$$

तब $F'(0)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a determinant whose entries are differentiable functions, $F'(x) = \Delta_1(x) + \Delta_2(x) + \Delta_3(x)$, where Δ_i is F with the i-th row replaced by its derivative and the other rows unchanged. All entries are differentiable on $(-1, 1)$.

At $x = 0$: $\sqrt{1} = 1$, $\sin 0 = 0$, $\log_e 1 = 0$; $\tan 0 = 0$, $e^0 = 1$, $0^2 = 0$; $1 - \cos 0 = 0$, $e^0 - 1 = 0$, $\dfrac{1}{1} = 1$.
So the rows at 0 are $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$: the matrix is the identity.

Derivatives of the rows: $R_1' = \left(\dfrac{1}{2\sqrt{1+x}}, \cos x, \dfrac{1}{1+x}\right)$, $R_2' = \left(\sec^2 x, \cos x\, e^{\sin x}, 2x\right)$, $R_3' = \left(\sin x, e^x, -\dfrac{1}{(1+x)^2}\right)$

At $x = 0$: $R_1'(0) = \left(\dfrac{1}{2}, 1, 1\right)$, $R_2'(0) = (1, 1, 0)$, $R_3'(0) = (0, 1, -1)$.

$$\Delta_1(0) = \begin{vmatrix} \frac{1}{2} & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{2} \text{ (upper triangular).}$$

$$\Delta_2(0) = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \text{ (lower triangular).}$$

$$\Delta_3(0) = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} = -1.$$

Hence $F'(0) = \dfrac{1}{2} + 1 - 1 = \dfrac{1}{2} = 0.50$. (Because the other two rows are rows of the identity, each $\Delta_i(0)$ is just the i-th diagonal entry of $R_i'(0)$.)

Final Answer: 0.50

Question 08

English

Let $p : \mathbb{R} \to \mathbb{R}$ be defined by $p(x) = (1+x)(1+2x)(1+3x)(1+4x)(1+5x)(1+6x)$. Then the value of $\lim\limits_{x\to 0} \dfrac{p(x) - 1 - 21x}{x^2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $p : \mathbb{R} \to \mathbb{R}$, $p(x) = (1+x)(1+2x)(1+3x)(1+4x)(1+5x)(1+6x)$ द्वारा परिभाषित है। तब $\lim\limits_{x\to 0} \dfrac{p(x) - 1 - 21x}{x^2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At $x = 0$: $p(0) = 1$, so the numerator tends to 0, as does x^2.

Logarithmic derivative: $\dfrac{p'(x)}{p(x)} = \sum\limits_{k=1}^{6} \dfrac{k}{1+kx}$, so $p'(0) = 1 + 2 + \cdots + 6 = 21$. Hence the derivative of the numerator, $p'(x) - 21$, also tends to 0.

By L'Hospital's rule twice (both stages are $\dfrac{0}{0}$ forms of differentiable functions): $\lim\limits_{x\to 0} \dfrac{p(x) - 1 - 21x}{x^2} = \lim\limits_{x\to 0} \dfrac{p'(x) - 21}{2x} = \lim\limits_{x\to 0} \dfrac{p''(x)}{2} = \dfrac{p''(0)}{2}$.

Differentiate $p' = pS$ with $S(x) = \sum\limits_{k=1}^{6} \dfrac{k}{1+kx}$: $p'' = p'S + pS' = p\left(S^2 + S'\right)$, and $S'(x) = -\sum\limits_{k=1}^{6} \dfrac{k^2}{(1+kx)^2}$.

At $x = 0$: $S(0) = 21$ and $S'(0) = -(1 + 4 + 9 + 16 + 25 + 36) = -91$, so $p''(0) = 441 - 91 = 350$.

Therefore the limit is $\dfrac{350}{2} = 175$. (This is the coefficient of x^2 in $p(x)$, the sum of the products jk over $1 \le j < k \le 6$.)

Final Answer: 175

Question 09

English

Let a and b be real numbers such that $L = \lim_{x \to 0} \dfrac{a \sin^{-1} x + b \tan x - \sin x}{x^5}$ exists and is finite, where $\sin^{-1}$ denotes the principal value of the inverse sine function. Then the value of $\dfrac{a}{L}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b ऐसी वास्तविक संख्याएँ हैं कि $L = \lim_{x \to 0} \dfrac{a \sin^{-1} x + b \tan x - \sin x}{x^5}$ का अस्तित्व है तथा यह परिमित है, जहाँ $\sin^{-1}$ प्रतिलोम ज्या फलन का मुख्य मान है। तब $\dfrac{a}{L}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The numerator $N(x) = a \sin^{-1} x + b \tan x - \sin x$ vanishes at 0, so for a finite limit we may apply L'Hospital's rule once (the new limit, if it exists, equals L): $L = \lim_{x \to 0} \dfrac{N'(x)}{5x^4}$ with $N'(x) = \dfrac{a}{\sqrt{1 - x^2}} + b \sec^2 x - \cos x$. Differentiating once replaces $\sin^{-1} x$ by an algebraic function, which is easier to expand; further L'Hospital steps would be cumbersome, so expand instead.

Maclaurin expansions up to x^4: $(1 - x^2)^{-1/2} = 1 + \dfrac{x^2}{2} + \dfrac{3x^4}{8} + O(x^6)$ (binomial series); $\tan x = x + \dfrac{x^3}{3} + O(x^5)$, so $\sec^2 x = 1 + \tan^2 x = 1 + x^2 + \dfrac{2x^4}{3} + O(x^6)$; $\cos x = 1 - \dfrac{x^2}{2} + \dfrac{x^4}{24} + O(x^6)$.

So $N'(x) = (a + b - 1) + \left(\dfrac{a}{2} + b + \dfrac{1}{2}\right) x^2 + \left(\dfrac{3a}{8} + \dfrac{2b}{3} - \dfrac{1}{24}\right) x^4 + O(x^6)$.

For $\dfrac{N'(x)}{5x^4}$ to have a finite limit, the constant term and the x^2 term must be zero: $a + b = 1$ and $a + 2b = -1$. Hence $b = -2$, $a = 3$.

Then the x^4 coefficient is $\dfrac{9}{8} - \dfrac{4}{3} - \dfrac{1}{24} = \dfrac{27 - 32 - 1}{24} = -\dfrac{1}{4}$, so $L = \dfrac{-1/4}{5} = -\dfrac{1}{20}$, which is finite and non-zero.

(Check by direct expansion: $3 \sin^{-1} x - 2 \tan x - \sin x = 3 \cdot \dfrac{3x^5}{40} - 2 \cdot \dfrac{2x^5}{15} - \dfrac{x^5}{120} + O(x^7) = -\dfrac{x^5}{20} + O(x^7)$.)

Therefore $\dfrac{a}{L} = \dfrac{3}{-1/20} = -60$.

Final Answer: -60

Question 10

English

Let $L = \lim\limits_{x\to 0^+} \left[(x-\sin x)\left(e^{2x}-1\right)^2\right]^{\frac{4}{\log_e(\tan x)}}$. Then the value of $\log_e L$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $L = \lim\limits_{x\to 0^+} \left[(x-\sin x)\left(e^{2x}-1\right)^2\right]^{\frac{4}{\log_e(\tan x)}}$ है। तब $\log_e L$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $0 < x < \dfrac{\pi}{4}$: $x - \sin x > 0$, $e^{2x} - 1 > 0$ and $0 < \tan x < 1$, so the base is positive and the expression is defined. As $x \to 0^+$ the base tends to 0 and $\log_e(\tan x) \to -\infty$, so the exponent tends to 0: the limit is of the form 0^0.

Taking logarithms: $\log_e L = \lim\limits_{x\to 0^+} \dfrac{4\log_e\left[(x-\sin x)\left(e^{2x}-1\right)^2\right]}{\log_e(\tan x)} = \lim\limits_{x\to 0^+} \dfrac{4\left[\log_e(x-\sin x) + 2\log_e\left(e^{2x}-1\right)\right]}{\log_e(\tan x)}$ (an $\dfrac{\infty}{\infty}$ form).

Standard limits: $\dfrac{x-\sin x}{x^3} \to \dfrac{1}{6}$, $\dfrac{e^{2x}-1}{2x} \to 1$, $\dfrac{\tan x}{x} \to 1$. Hence $\log_e(x-\sin x) = 3\log_e x + \log_e \dfrac{1}{6} + o(1)$, $\log_e\left(e^{2x}-1\right) = \log_e x + \log_e 2 + o(1)$ and $\log_e(\tan x) = \log_e x + o(1)$.

So the numerator is $4\left[5\log_e x + \log_e \dfrac{4}{6} + o(1)\right]$ and the denominator is $\log_e x + o(1)$.

Dividing numerator and denominator by $\log_e x$, which tends to $-\infty$, the constant terms vanish: $\log_e L = \dfrac{4\cdot 5}{1} = 20$.

So $L = e^{20}$ and $\log_e L = 20$. (Counting $x - \sin x$ as order x or x^2 instead of x^3 gives 12 or 16.)

Final Answer: 20